Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 1 of 13 · find part c in this mark scheme extract

3(a)	A1 A2 A3 A4	Award **one** mark for correct column (A OR B) Award **one** mark for correct column (NOT B) Award **one** mark for first set of two cells in third column Award **one** mark for second set of two cells in third column	Award follow through on last column only 5th column can be judged • On columns 1 and 2 for MP3 and MP4 or • as FT from columns 3 and 4 for MP3 and MP4 • award highest mark	4

A	B	A OR B	NOT B	(A OR B) AND (NOT B)
0	0	0	1	0
0	1	1	0	0
1	0	1	1	1
1	1	1	0	0

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 2 of 13 · find part c in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 3 of 13 · find part c in this mark scheme extract

3(b)		Award **one** mark for each of:		
	B1	Correct output taken from array (1)	Must be from array i.e. Indexing used	
	B2	Use of iteration to loop over every item in the array (1)	Award foreach loops Accept any array	
	B3	Use of appropriate method to step accurately, i.e. every other item starting at position 0 (1)	Any appropriate step method Not hard coded index	**3**

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 4 of 13 · find part c in this mark scheme extract

C#

```
// Q03bFINISHED

String[] starNames = {"Alasia", "Beid", "Castor", "Denebola",
                      "Electra", "Fafnir", "Gudja", "Haedus",
                        "Izar", "Jishui", "Kang", "Lich",
                        "Maia", "Nahn", "Ogma", "Peacock"};


// Write your code here
int start = 0;
int stop = starNames.Length;
int step = 2;

for (int index = start; index < stop; index = index + step)
{
    Console.WriteLine (starNames[index]);
}

// Alternative
for (int i=0; i<starNames.Length; i=i+2)
{
    Console.WriteLine (starNames[i]);
}

```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 5 of 13 · find part c in this mark scheme extract

Java

```java
// Q03bFINISHED


public class Q03bFINISHED
{
    public static void main (String[] args)
    {
        String starNames[] = {"Alasia", "Beid", "Castor", "Denebola",
                              "Electra", "Fafnir", "Gudja", "Haedus",
                              "Izar", "Jishui", "Kang", "Lich",
                              "Maia", "Nahn", "Ogma", "Peacock"};

        // Write your code here
        int start = 0;
        int stop = starNames.length;
        int step = 2;

        for (int index = start; index < stop; index = index + step)
        {
            System.out.println (starNames[index]);
        }

        // Alternative
        for (int i=0; i<starNames.length; i=i+2)
        {
            System.out.println (starNames[i]);
        }

    }
}
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 6 of 13 · find part c in this mark scheme extract

Python

```python
# Q03bFINISHED

starNames = ["Alasia", "Beid", "Castor", "Denebola",
             "Electra", "Fafnir", "Gudja", "Haedus",
             "Izar", "Jishui", "Kang", "Lich",
             "Maia", "Nahn", "Ogma", "Peacock"]

# Write your code here

start = 0
stop = len (starNames)
step = 2

for index in range (start, stop, step):
    print (starNames[index])

# Alternative
for index in range (0, len (starNames), 2):
    print (starNames[index])
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 7 of 13 · find part c in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 8 of 13 · find part c in this mark scheme extract

3(c)		Award **one** mark for each:		
	C1	Take a string input (1)	• `aString = Console.ReadLine ();` • `aString = myKeyboard.nextLine();` • `aString = input ("Enter a number: ")`	**6**
	C2	Complete the selection with 'if' (1)		
	C3	Convert string to integer (1)	• `aNum = Convert.ToInt32 (aString);` • `aNum = Integer.parseInt (aString);` • `aNum = int (aString)`	
	C4	Any one correct set of relational operators (1) ignore logical operator first two tests ignore test dealing with >= 1	Python • `(aNum <= 20) or (aNum >= 60)` • `(aNum < 21) or (aNum > 59)` • `(aNum > 30) and (aNum < 40)` • `(aNum >= 31) and (aNum <= 39)`	
	C5	Both logical operators used correctly in each test (1) Ignore relational operators First test: **or** (must attempt to address 20 and 60 with relational, even if C4 not awarded Second test: **and**	C# and Java • `((aNum <= 20) || (aNum >= 60))` • `(aNum < 21) || (aNum > 59))` • `((aNum > 30) && (aNum < 40))` • `((aNum >= 31) && (aNum <= 39))`	
	C6	Complete the third test (1)	All Languages: `(aNum == 30)`	

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 9 of 13 · find part c in this mark scheme extract

C#

```
// Q03cFINISHED

String aString = "";
int aNum = 0;

Console.WriteLine ("Enter a number:");

// Complete the line to take a string input
aString = Console.ReadLine ();

// Complete the selection statement
if (aString != "")
{
    // Convert the string to an integer
    aNum = Convert.ToInt32 (aString);
    if (aNum > 0)
    {
        // Complete the test
        if ((aNum <= 20) || (aNum >= 60))
        {
            Console.WriteLine ("Acceptable");
        }
        // Complete the test
        else if ((aNum >30) && (aNum < 40))
        {
            Console.WriteLine ("Centre");
        }
        // Complete the test
        else if (aNum == 30)
        {
            Console.WriteLine ("Perfect");
        }
        else
        {
            Console.WriteLine ("No message");
        }
    }
    else
    {
        Console.WriteLine ("The number must be greater than zero");
    }
}
else
{
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 10 of 13 · find part c in this mark scheme extract

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 11 of 13 · find part c in this mark scheme extract

Java

```java
// Q03cFINISHED

import java.util.Scanner;

public class Q03cFINISHED
{
    public static void main (String[] args)
    {
        String aString = "";
        int aNum = 0;
        Scanner myKeyboard = new Scanner (System.in);

        System.out.println ("Enter a number:");

        // Complete the line to take a string input
        aString = myKeyboard.nextLine();

        // Complete the selection statement
        if (aString != "")
        {
            // Convert the string to an integer
            aNum =  Integer.parseInt (aString);
            if (aNum > 0)
            {
                // Complete the test
                if ((aNum <= 20) || (aNum >= 60))
                {
                    System.out.println ("Acceptable");
                }
                // Complete the test
                else if ((aNum >30) && (aNum < 40))
                {
                    System.out.println ("Centre");
                }
                // Complete the test
                else if (aNum == 30)
                {
                    System.out.println ("Perfect");
                }
                else
                {
                    System.out.println ("No message");
                }
            }
            else
            {
                System.out.println ("The number must be greater than zero");
            }
        }
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 12 of 13 · find part c in this mark scheme extract

Python

```python
# Q03cFINISHED

# Complete the line to take a string input
aString = input ("Enter a number: ")

# Complete the selection statement
if (aString != ""):

    # Convert the string to an integer
    aNum = int (aString)

    if (aNum > 0):

        # Complete the test
        if (aNum <= 20) or (aNum >= 60):
            print ("Acceptable")
        # Complete the test
        elif (aNum > 30) and (aNum < 40):
            print ("Centre")
        # Complete the test
        elif (aNum == 30):
            print ("Perfect")
        else:
            print ("No message")
    else:
        print ("The number must be greater than zero")
else:
    print ("You must provide a number")
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 1 of 6 · Question 3 part c · 6 marks

2024 JUN · Paper 02 · extract 13 of 13 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 1 of 10 · find part f in this mark scheme extract

1(a)	**The only correct answer is C** A is not correct because Boolean only has values True and False B is not correct because it only allows for one character D is not correct because it is used for mixed text and numeric data		(1)

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 2 of 10 · find part f in this mark scheme extract

1(b)	(an error that) produces incorrect /unintended output (1)	Accept an example of an explained logic error	(1)

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 3 of 10 · find part f in this mark scheme extract

1(d)	Any one from: 1. explain how the program works (1) 2. improve readability of the program (1) 3. make code more understandable (1)		(1)

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 4 of 10 · find part f in this mark scheme extract

1(e)	1. Global variables created in the main program / can be accessed from any part of the program / are in scope throughout the program (1) 2. Local variables are declared/created inside a function/blocked code (indented, brackets) and are only accessible there / are only in scope within the function in which they're declared (1)		(2)

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 5 of 10 · find part f in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 6 of 10 · find part f in this mark scheme extract

1(g)	For up to **two** tests Award 1 mark for an appropriate validation test and 1 mark for an item of test data that would fail **only** the given test **Validation test** / **Erroneous data** Are the first two characters letters/upper case letters? / 1234CCC Are the third- and fourth-characters numerical digits? / CCCCBBB Are the last three characters letters/upper case letters? / CC12345	Each test should target a different feature of one of the requirements • Exactly 7 characters (in the stem) • First two characters are letters / upper case letters • Characters 3 & 4 are numeric • Last three characters are letters / upper case letters	(4)

Question	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 7 of 10 · find part f in this mark scheme extract

1(c)	**C#** 1. Semicolon at the end of line 9 [maxValue=5;] (1) 2. Consistent capitalisation of 'count' on lines 10, 11, 13 and 14 (1) 3. constantValue needs to be defined and given the value 7 / replaced by value 7 (1) **Java** 1. Semicolon at the end of line 8 [maxValue = 5;] (1) 2. Consistent capitalisation of 'count' on lines 7, 9, 11 and 12 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1) **Python** 1. Colon needed at end of line 7 [count, maxValue):] (1) 2. Consistent capitalisation of 'count' on line 3, 7, 8 and 9 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1)	Candidates are required to open the file Q01c in the code editor. Amended code should be saved as Q01cFINISHED. Line numbers may have been changed by corrections to code (e.g. insertion of constantValue definition)	(3)

Question	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 8 of 10 · find part f in this mark scheme extract

<table>
<tr>
<td>1(f)</td>
<td>Award 1 mark for each correct pair of relational operators and 1 mark for correct Boolean operator for each condition.

<table>
<tr><th>Condition</th><th>Coding</th></tr>
<tr><td rowspan="2">Shop income more than £5000 or assistant sold at least 10 pairs</td><td>shopIncome > 5000; assistantSales >= 10 (1)</td></tr>
<tr><td>correct OR operator (1)</td></tr>
<tr><td rowspan="2">Shop income at least £2000 and assistant sold at least 5 pairs</td><td>shopIncome >= 2000; assistantSales >= 5 (1)</td></tr>
<tr><td>correct AND operator (1)</td></tr>
</table>

Accept >9, >1999 and >4 for respective >= comparisons</td>
<td>Candidates are required to open the file Q01f in the code editor.
Amended code should be saved as Q01fFINISHED.
Do not penalise candidates who attempt more than the stated requirements.
Do not award Boolean operator mark for single | or & as these are not correct.</td>
<td>(4)</td>
</tr>
</table>

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 9 of 10 · find part f in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 2 of 6 · Question 1 part f · 4 marks

2019 JUN · Paper 02 · extract 10 of 10 · find part f in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 3 of 6 · Question 4 part a · 6 marks

2020 NOV · Paper 02 · extract 1 of 4 · find part a in this mark scheme extract

4 (a)	A1	2.5.2	english < 40, maths < 50 (1)		
	A2	2.5.3	Use of **and** operator in IF (1)		
	A3	2.5.2	english < 40, maths < 50 (1)		
	A4	2.5.3	Use of **or** operator in 'else if 1' (1)		
	A5	2.5.2	english >= 80 AND maths >=85 (1)		
	A6	2.5.3	Correct completion of final else with messageIndex = 3 (1)		(6)
Code example					
Python					

```
if (english < 40 and maths < 50):
    messageIndex = 0
elif (english < 40 or maths < 50):
    messageIndex = 1
elif (english >= 80 and maths >= 85):
    messageIndex = 2
else:
    messageIndex = 3
```

12 of 16

Type MS · Finish an if/elif/else so each input gives the right message

Answer 3 of 6 · Question 4 part a · 6 marks

2020 NOV · Paper 02 · extract 2 of 4 · find part a in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 3 of 6 · Question 4 part a · 6 marks

2020 NOV · Paper 02 · extract 3 of 4 · find part a in this mark scheme extract

4 (b)	B1	2.2.2	Loop checks each pupil for low attendance		(7)
	B2	2.1.6	Display of name of low attendances		
	B3	2.3.1	Counter for high attenders initialised		
	B4	2.2.2	Loop uses correct comparison (>=)		
	B5	2.6.2	Number of high attenders displayed correctly		
	B6	2.6.3	Subprogram for option 1 called correctly		
	B7	2.6.3	Subprogram for option 2 called correctly		

Code example

Python

```
#   write subprogram for menu option 1 here
def lowAttendance():
    for x in range(len(pupilAttendance)):
        if pupilAttendance[x-1][2] < 75 :
            print (pupilAttendance[x-1][0], pupilAttendance[x-1][1])
    return

#   write subprogram for menu option 2 here

def highAttendance():
    count = 0
    for x in range(len(pupilAttendance)):
        if pupilAttendance[x-1][2] >= 90 :
            count += 1
    return count
```

```
Faroukh Salah
Amara Grzinski
Taz Grimstow
Sadia Bhatti
Fernado Askabat
Siyao Wang
```

```
if option == 1:
    # complete if statement
    lowAttendance()

elif option == 2:
    # complete else if statement
    highAtt = highAttendance()
    print ("There were ", highAtt," high attenders")
```

```
There were  7  high attenders
```

13 of 16

Type MS · Finish an if/elif/else so each input gives the right message

Answer 3 of 6 · Question 4 part a · 6 marks

2020 NOV · Paper 02 · extract 4 of 4 · find part a in this mark scheme extract

For Q6, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 4 of 6 · Question 2 part b · 4 marks

2021 JUN · Paper 02 · extract 1 of 6 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark
2 (c)	Award 1 mark for each set of test data.		(3)

		Test data	Expected results
C1 C2 C3	booksSold profit	Either of • booksSold = 4 • profit = 4	Poor performances this week
	booksSold	5	Sales and profit are good this week
	profit	10	
	booksSold	21	Sales and profit are excellent this week
	profit	20	

Type MS · Finish an if/elif/else so each input gives the right message

Answer 4 of 6 · Question 2 part b · 4 marks

2021 JUN · Paper 02 · extract 2 of 6 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 4 of 6 · Question 2 part b · 4 marks

2021 JUN · Paper 02 · extract 3 of 6 · find part b in this mark scheme extract

Question		Answer	Additional Guidance	Mark
2 (a)	Award one mark for each of:		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the assignment of username and/or password Do not penalise spelling mistakes and alternative wording of the output.	(11)
	A1	At least one variable with a suitable variable name		
	A2	username = bard423		
	A3	password = nX2934?		
	A4	Loop used		
	A5	Username or password entered		
	A6	Username or password stored in variable(s)		
	A7	At least one suitable input message		
	A8	Checks username and password		
	A9	Appropriate error message(s) displays		
	A10	Welcome message displayed		
	A11	Executing and producing correct output (must have A4)		
Code example				
Python				

```
# Initialise variables
username = "bard423"
password = "nX2934?"
count = 0
inputUsername = ""
inputPassword = ""

# Print promts, take and check input from user
while inputUsername != username or inputPassword != password:
    if count > 0:
        print("There is a problem with the login details. Try again")
    count = count + 1
    inputUsername = input("Enter your username ")
    inputPassword = input("Enter your password ")
print("Welcome")
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 4 of 6 · Question 2 part b · 4 marks

2021 JUN · Paper 02 · extract 4 of 6 · find part b in this mark scheme extract

Type MS · Finish an if/elif/else so each input gives the right message

Answer 4 of 6 · Question 2 part b · 4 marks

2021 JUN · Paper 02 · extract 5 of 6 · find part b in this mark scheme extract

Question	Mp	Answer		Additional Guidance	Mark
2 (b)	Award 1 mark for each correct condition.			Accept alternatives e.g. Line 11 booksSold <=4 etc.	(4)
		Condition	**Output message**		
	B1	Number of books sold is under 5 or profit made is under 5	Poor performance this week		
	B2	Number of books sold is over 20; profit made is at least 20	Sales and profit are excellent this week		
	B3	Number of books sold is at least 5; profit made is at least 10	Sales and profit are good this week		
	B4	All other inputs	Alert manager		
Code example					
Python					

```
11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
15 elif booksSold >=5 and profit >= 10:
16     print("Sales and profit are good this week")
17 else:
18     print("Alert manager")
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 4 of 6 · Question 2 part b · 4 marks

2021 JUN · Paper 02 · extract 6 of 6 · find part b in this mark scheme extract

Type MS · Finish an if/elif/else so each input gives the right message

Answer 5 of 6 · Question 1 part g · 4 marks

2022 JUN · Paper 02 · extract 1 of 9 · find part g in this mark scheme extract

1 (a)	A1	**The only correct answer is D** A is not correct because the value can be used more than once B is not correct because the value does not always have to be input C is not correct because the value does not always have to be used in a calculation		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 5 of 6 · Question 1 part g · 4 marks

2022 JUN · Paper 02 · extract 2 of 9 · find part g in this mark scheme extract

1 (b)	B1	**The only correct answer is B** A is not correct because this would not aid readability of the code C is not correct because this would not aid readability of the code D is not correct because this would not aid readability of the code		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 5 of 6 · Question 1 part g · 4 marks

2022 JUN · Paper 02 · extract 3 of 9 · find part g in this mark scheme extract

1 (c)	C1 C2	**Award 1 mark for each of:** **Data type** / **Example value** char / An example of any single letter, number or symbol real / An example of a number with a decimal point	Accept quotes around the character Real number does not need a number after the decimal point to be awarded the mark	(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 5 of 6 · Question 1 part g · 4 marks

2022 JUN · Paper 02 · extract 4 of 9 · find part g in this mark scheme extract

1 (d)	D1	**Award up to 2 marks for a description such as:** • Boundary testing uses the most extreme valid data / minimum and maximum valid values (1) whereas erroneous testing uses invalid data (1)		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 5 of 6 · Question 1 part g · 4 marks

2022 JUN · Paper 02 · extract 5 of 9 · find part g in this mark scheme extract

1 (e)	E1	Logic/logical		(1)

Type MS · Finish an if/elif/else so each input gives the right message

Answer 5 of 6 · Question 1 part g · 4 marks

2022 JUN · Paper 02 · extract 6 of 9 · find part g in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 5 of 6 · Question 1 part g · 4 marks

2022 JUN · Paper 02 · extract 7 of 9 · find part g in this mark scheme extract

1 (f)	**Award 1 mark for each of:**			
	F1	Bracket added and no other amendments made to the line (1)		
	F2	== becomes = (1)		
	F3	" moved to after **is** and before , OR , area changed to +str(area) (1) Must still include the print statement		(3)
Code examples				
Python	3 length = float(input("Enter the first number ") 4 5 area = length * length 6 7 print("The area of the square is" , area)			

Question	mp	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 5 of 6 · Question 1 part g · 4 marks

2022 JUN · Paper 02 · extract 8 of 9 · find part g in this mark scheme extract

1 (g)	**Award 1 mark for each of:**			
	G1	At least 1 condition correct (1)		
	G2	At least 1 > or < condition and message match (1)		
	G3	At least 1 output includes letter and stored letter with an appropriate message (1)		
	G4	All conditions and outputs correct (1)		(4)
Code examples				
Python				

```
# Amend the code by completing the if statement
if letter > storedLetter:
    print(letter + " is later in the alphabet than " + storedLetter)
elif letter == storedLetter:
    print(letter + " is the same as " + storedLetter)
else:
    print(letter + " is earlier in the alphabet than " + storedLetter)
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 5 of 6 · Question 1 part g · 4 marks

2022 JUN · Paper 02 · extract 9 of 9 · find part g in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 1 of 16 · find part b in this mark scheme extract

1(a)	A1	**The only correct answer is D** ***A*** *is not correct because it is the definition of decomposition* ***B*** *is not correct because it is a task undertaken during evaluation* ***C*** *is not correct because it describes a method for expressing algorithms*		**1**

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 2 of 16 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 3 of 16 · find part b in this mark scheme extract

1b		Award **one** mark for each of:		
	B1	Initialisation of myName to "" (1)	• `String myName = "";` • `String myName = "";` • `myName = ""` Allow null Allow single space between quotes " "	
	B2	Add a line to create myAge and initialise it to 0 (1)	• `int myAge = 0;` • `int myAge = 0;` • `myAge = 0` Allow any method	
	B3	Complete the test by inserting < (1)	• `if (myAge < 30)` • `if (myAge < 30)` • `if (myAge < 30):` Allow <= Can award reverse test only if welcome message contained within an 'else' part	
	B4	Add a line to display "Welcome" (1)	• `Console.WriteLine ("Welcome");` • `System.out.println ("Welcome");` • `print ("Welcome")` ignore indentation Allow any method	**4**

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 4 of 16 · find part b in this mark scheme extract

C#

```
// Q01bFINISHED

// Complete the line to initialise myName to an empty string
String myName = "";

// Add a line to create an integer variable called myAge and
//     initialise myAge to 0
int myAge = 0;

Console.WriteLine ("Enter your name: ");
myName = Console.ReadLine();
Console.WriteLine ("Enter your age: ");
myAge = Convert.ToInt32 (Console.ReadLine());

// Complete the test to check if the age is less than 30
if (myAge < 30)
{
    // Add a line to display a message that says "Welcome"
    Console.WriteLine ("Welcome");
}
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 5 of 16 · find part b in this mark scheme extract

Java

```java
// Q01bFINISHED

import java.util.Scanner;
public class Q01bFINISHED
{
    public static void main (String[] args)
    {
        Scanner myKeyboard = new Scanner (System.in);

        // Complete the line to initialise myName to an empty string
        String myName = "";

        // Add a line to create an integer variable called myAge and
        //     initialise myAge to 0
        int myAge = 0;

        System.out.println ("Enter your name: ");
        myName = myKeyboard.nextLine();
        System.out.println ("Enter your age: ");
        myAge = Integer.parseInt (myKeyboard.nextLine());

        // Complete the test to check if the age is less than 30
        if (myAge < 30)
        {
            // Add a line to display a message that says "Welcome"
            System.out.println ("Welcome");
        }
    }
}
```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 6 of 16 · find part b in this mark scheme extract

Python

```python
# Q01bFINISHED

# Complete the line to initialise myName to an empty string
myName = ""

# Add a line to create an integer variable called myAge and
#     initialise myAge to 0
myAge = 0

myName = input ("Enter your name: ")
myAge = int (input ("Enter your age: "))

# Complete the test to check if the age is less than 30
if (myAge < 30):

    # Add a line to display a message that says "Welcome"
    print ("Welcome")

```

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 7 of 16 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 8 of 16 · find part b in this mark scheme extract

1c(i)		Award **one** mark for any of the following:		
	C1	C#: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Java: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Python: • # Q01c (1) • Q01c (1) • # Loop for 50 times (1) Loop for 50 times (1)	Ignore transcription errors	**1**

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 9 of 16 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 10 of 16 · find part b in this mark scheme extract

1c(ii)		Award **one** mark for any of the following:		
	C2	C#: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Java: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Python: • count = 0 (1) • total = 0 (1) • subtotal = 0 (1)	Ignore transcription errors	1
Question	**mp**	**Answer**	**Additional guidance**	**Mark**

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 11 of 16 · find part b in this mark scheme extract

1c(iii)		Award **one** mark for:		
	C3	• if (1)	Applies to all programming languages Ignore capitalisation	**1**

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 12 of 16 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 13 of 16 · find part b in this mark scheme extract

1c(iv)		Award **one** mark for:		
	C4	C#: • && (1) Java: • && (1) Python: • and (1)	Ignore capitalisation	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 14 of 16 · find part b in this mark scheme extract

1(d)	D1	**The only correct answer is B** ***A*** *is not correct because the data can be stored in an array as it is all the same data type (string)* ***C*** *is not correct because the data can be stored in an array as it is all the same data type (integer)* ***D*** *is not correct because the data can be stored in an array as it is all the same data type (real)*		**1**

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 15 of 16 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Finish an if/elif/else so each input gives the right message

Answer 6 of 6 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 16 of 16 · find part b in this mark scheme extract

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<tr>
<td>1(e)</td>
<td>E1
E2</td>
<td>One mark for each correct cell:
<table>
<tr><th>Item</th><th>Data type</th></tr>
<tr><td>45.82</td><td>All programming languages:
real/float
C#:
decimal/double
Java:
Double/double/BigDecimal
Python:
decimal</td></tr>
<tr><td>True</td><td>All programming languages:
Boolean/bool</td></tr>
</table></td>
<td>Ignore capitalisation
Do not accept string for True</td>
<td>2</td>
</tr>
</table>

Question	mp	Answer	Additional Guidance	Mark